EPFL - Computer Science

CS-401 Applied Data Analysis

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Introduction to ADA

Data analysis

"... the process of inspecting, cleaning, transforming, and modeling data with the goal of discovering useful information, suggesting conclusions, and supporting decision-making."

"Data analysis has multiple facets and approaches, encompassing diverse techniques under a variety of names, in different business, science, and social science domains."

Applied Data Analysis

- This course is about **breadth**, not depth
- "What methods, principles, and tools are out there?, rather than "How can I become an expert in deep learning for computer vision applied to images of cats?"
- Data science is a fast-paced, shifting fields
- Obsessing on one tool or technique won't pay off in a few years
- Be ready to explore and keep learning on your own

Goal: Enable you to conduct a full-fledge data science project from start to finish

Syllabus

- Handling data
  - "Slicing and dicing": obtaining, preparing, juggling data
- Visualizing data
  - Exploration of data, communication of results
- Describing data
  - How to support (and be suspicious of) claims about data
- Regression analysis for disentangling data
  - How to disentangle datasets with correlated variables
- Casual analysis of observational data

  - How to deal with "found data"
  - correlation != causation

Syllabus (cont'd)

- Learning from data
  - supervised learning
  - unsupervised learning
  - applied aspects of machine learning
- Handling specific types of data
  - handling text data
  - handling network data
- Scaling to massive data

Grading

- 15% Homework assignment
- 50% Final Exam
- 35% Project